and expressing the planet's mass in terms of its density ρ as $m = (4\pi/3)r^3\rho$, and performing some algebra, we obtain $M < \sqrt{3}c^3/\sqrt{2\pi G^3\rho}$. I estimate the density of Miller's planet to be ρ = 10,000 kilograms/meter^3 (about that of compressed rock), from which I obtain $M < 3.4 \times 10^{38}$ kilograms for Gargantua's mass, which is about the same as 200 million suns—which in turn I approximate as 100 million suns.

Using Einstein's relativistic equations, I have deduced a formula that connects the slowing of time on Miller's planet, S = one hour/(seven years) = 1.63×10^{-5} to the fraction α by which Gargantua's spin rate is less than its maximum possible spin: $\alpha = 16S^3/(3\sqrt{3})$. This formula is correct only for very fast spins. Inserting the value of S, we obtain $\alpha = 1.3 \times 10^{-14}$; that is, Gargantua's actual spin is less than its maximum possible spin by about one part in a hundred trillion.

Chapter 8. Imaging Gargantua

The equations that I gave to Oliver James at Double Negative, for the orbital motion of light rays around Gargantua, are a variant of those in Appendix A of Levin and Perez-Giz (2008). Our equations for the evolution of bundles of rays are a variant of those in Pineult and Roeder (1977a) and Pineult and Roder (1977b). In several papers that we'll make available at http://arxiv.org/find/gr-qc, Paul Franklin's team and I give the specific forms of our equations and discuss details of their implementation and the simulations that resulted.

Chapter 12. Gasping for Oxygen

Here are the calculations that underlie my statements in Chapter 13. They are a nice example of how a scientist makes estimates. These numbers are very approximate; I quote them accurate to only one digit.